Math 110 — Week 8

Generalized eigenvectors and Jordan form (Notes Ch. 14–15, LADR 8A–8C, HW8)

§1 Facts used from earlier weeks

V is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Shorthand: "S kills v" means $Sv = 0$, and "q kills T" means $q(T) = 0$.

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

Why.

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: T of any vector is a combination of the Tu_i (which are 0) and the Tv_j.
3. Independent: if $\sum c_j Tv_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the u_i, and independence of the basis of V gives all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**

Why. $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n$ (T1) $\iff \operatorname{range} T = V$ (T3).

(T3) **Dimension shortcuts.**

(a) An independent list of n vectors is a basis.
(b) A subspace of dimension n is all of V.
(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.
(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Why.

1. (a), (b): an independent list extends to a basis, and every basis has length n.
2. (c): a basis of U followed by a basis of W is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length n, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis x_i, u_j of U and, separately, to a basis x_i, w_l of W. Together the x_i, u_j, w_l span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ (x, u, w combinations of the x_i, u_j, w_l), then $w = -x - u \in U \cap W$, so w is a combination of the x_i. The basis x_i, w_l of W is independent, so $w = 0$. Then $x + u = 0$, and independence of x_i, u_j gives $x = u = 0$.

(T4) **Products of operators.** If S, T are injective, so is ST. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

Why. $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in T commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are T-invariant.

Why. Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda)=0 \iff (z-\lambda) \mid p$.

Why. Divide p by $z-\lambda$: the remainder is a constant, and evaluating at λ shows it equals $p(\lambda)$.

(T7) **Eigenvalues.**

(a) λ is an eigenvalue $\iff$ $T-\lambda I$ is not injective.

(b) If $Tv=\lambda v$, then $p(T)v=p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

Why.

1. (a) is the definition: $Tv=\lambda v$ with $v\neq 0$ means $(T-\lambda I)v=0$ with $v\neq 0$.
2. (b): $T^k v=\lambda^k v$ by induction; add up the terms of p.
3. (d): take $v\neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v=0$ for some $p\neq 0$; p is not constant since $v\neq 0$.
4. Factor $p=c(z-\lambda_1)\cdots(z-\lambda_m)$ (T6). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v\neq 0$ to 0, so some $T-\lambda_j I$ is not injective (T4).

(T8) **Upper-triangular matrices.** $\mathcal{M}(T)$ is upper-triangular in $v_1,\ldots,v_n \iff$ every $\operatorname{span}(v_1,\ldots,v_j)$ is invariant. Then the eigenvalues of T are the diagonal entries. Over $\mathbb{C}$ every operator has an upper-triangular matrix.

Why.

1. Column j holds the coordinates of Tv_j, so "upper-triangular" means $Tv_j \in \operatorname{span}(v_1,\ldots,v_j)$ for every j.
2. If $a_{jj}=\lambda$: $T-\lambda I$ maps $\operatorname{span}(v_1,\ldots,v_j)$ into $\operatorname{span}(v_1,\ldots,v_{j-1})$, a smaller space, so it is not injective (T1). So λ is an eigenvalue.
3. If no a_{jj} equals λ: $\mathcal{M}(T-\lambda I)$ is triangular with nonzero diagonal. Its columns are independent: in a relation, the right-most column used is the only one with a nonzero entry in its own diagonal row. So $T-\lambda I$ is injective.
4. Existence over $\mathbb{C}$, by induction on n: take an eigenvalue λ (T7(d)). $U=\operatorname{range}(T-\lambda I)$ is invariant (T5) and $\dim U<n$, since $T-\lambda I$ is not injective (T1).
5. By induction U has a basis $u_1,\ldots,u_m$ with every $\operatorname{span}(u_1,\ldots,u_j)$ invariant. Extend it to a basis $u_1,\ldots,u_m,v_1,\ldots,v_k$ of V.
6. $Tv_j=(T-\lambda I)v_j+\lambda v_j \in U+\operatorname{span}(v_j) \subseteq \operatorname{span}(u_1,\ldots,u_m,v_1,\ldots,v_j)$. So every span of an initial part of the list is invariant.

(T9) **Division with remainder.** For polynomials p and $s\neq 0$ there are unique q,r with $p=sq+r$ and $r=0$ or $\deg r<\deg s$.

Why.

1. Existence, by induction on $\deg p$: if $\deg p<\deg s$, take $q=0$, $r=p$. Otherwise subtract $\frac{a}{b}x^{\deg p-\deg s}s$ from p (a,b the leading coefficients of p,s); this lowers the degree, so induction applies.
2. Uniqueness: if $sq+r=sq'+r'$, then $s(q-q')=r'-r$. The right side has degree $<\deg s$, so $q=q'$ and $r=r'$.

(T10) **Block diagonal matrices.** If $V=U\oplus W$ with both T-invariant, then in a basis of U followed by one of W, $\mathcal{M}(T)=\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T=\operatorname{null} T|_U \oplus \operatorname{null} T|_W$.

Why. $Tu\in U$ is a combination of the U-basis only, so its column is 0 in the W-rows (same for W). If $T(u+w)=0$, then $Tu=-Tw\in U\cap W=\{0\}$.

(T11) **Change of basis.** If column j of S holds the B-coordinates of the j-th vector of B', then $\mathcal{M}_{B'}(T)=S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.

Why. S converts B'-coordinates to B-coordinates: apply S, then $\mathcal{M}_B(T)$, then convert back with S^{-1}. Every invertible S arises this way, since its columns form a basis.

(T12) **Quotient operator.** If U is T-invariant, then $(T/U)(v+U)=Tv+U$ is a well-defined operator on V/U, $\dim V/U=\dim V-\dim U$, and $p(T/U)(v+U)=p(T)v+U$.

Why.

1. Well defined: if $v+U=v'+U$, then $v-v'\in U$, so $Tv-Tv'\in U$ (U is invariant).

2. Dimension: $v \mapsto v + U$ maps V onto V/U with null space U; apply (T1).
3. $(T/U)^k(v+U) = T^k v + U$ by induction; add up the terms of p.

(T13) **Minimal polynomial (Week 7).** m_T is the monic polynomial of least degree with $m_T(T) = 0$. (a) $q(T) = 0 \iff m_T \mid q$ (Prop. 65). (b) The roots of m_T in $\mathbb{F}$ are exactly the eigenvalues (Prop. 66). (c) $\deg m_T \le n$ (Prop. 67). (d) If U is invariant, $m_{T|_U} \mid m_T$. (e) If $V = U_1 \oplus \cdots \oplus U_r$ with every U_i invariant, m_T is the least common multiple of the $m_{T|_{U_i}}$.

Why. (a) Divide q by m_T; the remainder is 0 at T and has smaller degree, so it is 0. (d) $m_T(T|_U)u = m_T(T)u = 0$, then (a). (e) $q(T) = 0$ exactly when $q(T|_{U_i}) = 0$ for all i, i.e. when every $m_{T|_{U_i}}$ divides q. (b), (c) are proved in the Week 7 handout.

§2 This week

§2.1 Generalized eigenvectors (Ch. 14)

The question. We want the simplest matrix for every operator. With an eigenbasis it is diagonal, but often there are too few eigenvectors.

- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has eigenvectors only on $\operatorname{span}(e_1)$.
- But $Ne_2 = e_1 \neq 0$ and $N^2 e_2 = 0$: e_2 is sent to 0 by N^2 but not by N.
- Idea of the chapter: allow $(T - \lambda I)^k v = 0$ for some $k \ge 1$ instead of only $k = 1$. Over $\mathbb{C}$ that gives enough vectors for a basis.

Definition (Def. 37 / LADR 8.8, 8.19)**.** $v \neq 0$ is a **generalized eigenvector** for λ if $(T - \lambda I)^k v = 0$ for some $k \ge 1$. The **generalized eigenspace** is $G(\lambda, T) = \bigcup_{k \ge 0} \operatorname{null}(T - \lambda I)^k$.

- $E(\lambda, T) \subseteq G(\lambda, T)$ (take $k = 1$). In the example, $E(0, N) = \operatorname{span}(e_1)$ but $G(0, N) = \mathbb{F}^2$.
- A union over all k is awkward. Props. 69–70 show it stops at a fixed power.
- An operator N is **nilpotent** if $N^k = 0$ for some k (Def. 39), e.g. N above. So $G(\lambda, T)$ is the part of V on which $T - \lambda I$ acts nilpotently.

$$v \xrightarrow{T-\lambda I} (T-\lambda I)v \xrightarrow{T-\lambda I} \cdots \xrightarrow{T-\lambda I} w \xrightarrow{T-\lambda I} 0$$

generalized eigenvector — last nonzero vector: an eigenvector

Repeatedly applying $T - \lambda I$ to a generalized eigenvector reaches 0. The last nonzero vector w has $(T - \lambda I)w = 0$, so it is an eigenvector (Prop. 71). Smallest case: $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_2 \mapsto e_1 \mapsto 0$.

Proposition (Prop. 69 / LADR 8.1, 8.2: the null-space chain)
$\{0\} = \operatorname{null} T^0 \subseteq \operatorname{null} T \subseteq \operatorname{null} T^2 \subseteq \cdots$. If $\operatorname{null} T^m = \operatorname{null} T^{m+1}$, then $\operatorname{null} T^m = \operatorname{null} T^{m+k}$ for all $k \ge 0$.

The idea

(a) Each null space contains the one before. (b) Once two neighbours are equal, all later ones are equal.

(a) A vector killed by T^k stays killed by $T^{k+1} = T \cdot T^k$.

Only one equality, $\operatorname{null} T^m = \operatorname{null} T^{m+1}$. To use it on a vector $v \in \operatorname{null} T^{m+k+1}$, we need some vector in $\operatorname{null} T^{m+1}$ built from v.

Split the power: $T^{m+k+1}v = T^{m+1}(T^k v)$. So $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$, which says $T^{m+k}v = 0$. That is the equality one step higher; repeat.

Example: The shift on $\mathbb{F}^3$ ($e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$) has $\dim \operatorname{null} T^k = 0, 1, 2, 3, 3, 3, \ldots$: it grows until two neighbours are equal, and after that all are equal.

Proof. 1. If $T^k v = 0$, then $T^{k+1} v = T(0) = 0$. So $\operatorname{null} T^k \subseteq \operatorname{null} T^{k+1}$.
2. Suppose $\operatorname{null} T^m = \operatorname{null} T^{m+1}$ and $T^{m+k+1} v = 0$. Then $T^k v \in \operatorname{null} T^{m+1} = \operatorname{null} T^m$.
3. So $T^{m+k} v = T^m (T^k v) = 0$. Hence $\operatorname{null} T^{m+k+1} \subseteq \operatorname{null} T^{m+k}$, and with step 1 they are equal. Induct on k.

□

Proposition (Prop. 70 / LADR 8.3, 8.20)
$\operatorname{null} T^n = \operatorname{null} T^{n+1}$. Hence $G(\lambda, T) = \operatorname{null}(T - \lambda I)^n$, a T-invariant subspace.

The idea

The chain has stopped by the power n.

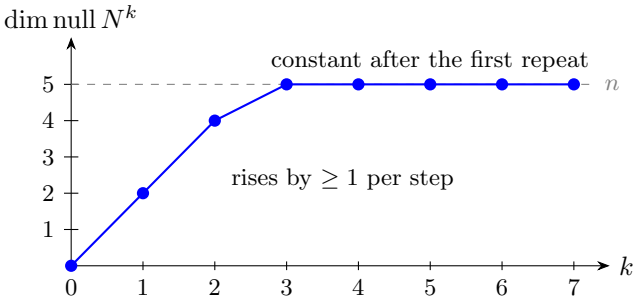
Until it stops, each step raises the dimension by at least 1 (a strictly larger subspace has larger dimension). Dimensions are at most n. Once it stops, it stays stopped (Prop. 69).

Count: starting from dimension 0, $n+1$ strict steps would reach dimension $n+1 > n$. So one of the first $n+1$ steps is an equality, and from there on the chain is constant.

Why it matters: The union defining $G(\lambda, T)$ is then the single null space $\operatorname{null}(T - \lambda I)^n$. So it is a subspace, and it is invariant by (T5).

Proof. 1. Suppose $\operatorname{null} T^n \neq \operatorname{null} T^{n+1}$. By Prop. 69 no earlier pair is equal either.
2. Then $0 = \dim \operatorname{null} T^0 < \dim \operatorname{null} T^1 < \cdots < \dim \operatorname{null} T^{n+1} \leq n$: that is $n+2$ distinct integers in $\{0, \ldots, n\}$. Impossible.
3. So $\bigcup_k \operatorname{null}(T - \lambda I)^k = \operatorname{null}(T - \lambda I)^n$. It is invariant by (T5).

□



N with blocks $J_3(0), J_2(0)$ on $\mathbb{F}^5$: $\dim \operatorname{null} N^k = 0, 2, 4, 5, 5, \ldots$ While the chain grows it gains at least 1 per step, and it cannot pass n, so it has stopped by $k = n$ (Prop. 70). Once two neighbours agree, all later ones agree (Prop. 69).

Proposition (LADR 8.4, 8.5: splitting off the nilpotent part)
$V = \operatorname{null} T^n \oplus \operatorname{range} T^n$, and T is injective on $\operatorname{range} T^n$.

The idea

Split V into a part where T is nilpotent ($\operatorname{null} T^n$) and a part where T is invertible ($\operatorname{range} T^n$). Applied to $T - \lambda I$, this splits off $G(\lambda, T)$; that is the first step of Prop. 73.

$\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1). So by (T3) it is enough to show $\operatorname{null} T^n \cap \operatorname{range} T^n = \{0\}$.

The obvious first attempt: Power 1: is $\operatorname{null} T \cap \operatorname{range} T = \{0\}$? No. For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $e_1 = N e_2$ is in the range and $N e_1 = 0$. A nonzero vector of range N can lie in null N.

With the power n, $\operatorname{range} T^n$ meets $\operatorname{null} T^n$ only in 0, because the chain has stopped. If $v = T^n u$ and $T^n v = 0$, then $T^{2n} u = 0$, and $\operatorname{null} T^{2n} = \operatorname{null} T^n$ (Prop. 70) gives $T^n u = 0$, i.e. $v = 0$.

Proof. 1. Let $v = T^n u$ with $T^n v = 0$. Then $T^{2n} u = 0$, so $u \in \operatorname{null} T^{2n} = \operatorname{null} T^n$, so $v = 0$.
2. $\dim \operatorname{null} T^n + \dim \operatorname{range} T^n = n$ (T1), so $V = \operatorname{null} T^n \oplus \operatorname{range} T^n$ (T3).
3. If $w \in \operatorname{range} T^n$ and $Tw = 0$, then $w \in \operatorname{null} T \subseteq \operatorname{null} T^n$, so $w = 0$.

□

Proposition (Prop. 71)
$G(\lambda, T) \neq \{0\} \iff \lambda$ is an eigenvalue of T.

The idea

A nonzero generalized eigenvector for λ forces an eigenvector for λ.

Apply $T - \lambda I$ to v repeatedly: $v \mapsto (T - \lambda I)v \mapsto \cdots \mapsto 0$. The last nonzero vector in this list is sent to 0 by $T - \lambda I$, so it is an eigenvector.

Proof. If λ is an eigenvalue, $\{0\} \neq E(\lambda, T) \subseteq G(\lambda, T)$. Conversely, let $v \in G(\lambda, T)$, $v \neq 0$, and let $j \geq 0$ be the largest exponent with $w = (T - \lambda I)^j v \neq 0$ (it exists because $(T - \lambda I)^k v = 0$ for some k). Then $(T - \lambda I)w = 0$ and $w \neq 0$. □

Proposition (Prop. 72 / LADR 8.12)
Generalized eigenvectors $v_1, \ldots, v_m$ for distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ are independent.

The idea

From $\sum a_i v_i = 0$, get each $a_k = 0$.

How independence of eigenvectors is proved (Prop. 54): apply an operator that kills every term except the k-th and is nonzero on the k-th. What is left is $a_k \cdot (\text{nonzero vector}) = 0$.

The obvious first attempt: $(T - \lambda_i I)^n$ kills v_i, so $S = \prod_{i \neq k} (T - \lambda_i I)^n$ kills every v_i with $i \neq k$. But it is not clear what S does to v_k, which is only a generalized eigenvector.

On an actual eigenvector w for λ_k, S is easy: $Sw = \prod_{i \neq k} (\lambda_k - \lambda_i)^n w \neq 0$ (T7). So first turn v_k into an eigenvector by applying $(T - \lambda_k I)^N$, as in Prop. 71. The extra factor still gives $Sv_i = 0$ for $i \neq k$, because all the factors commute.

Proof. 1. Fix k. Let N be the largest exponent with $w = (T - \lambda_k I)^N v_k \neq 0$. Then $Tw = \lambda_k w$, so $(T - \lambda_i I)^n w = (\lambda_k - \lambda_i)^n w$.
2. Apply $S = \prod_{i \neq k} (T - \lambda_i I)^n \cdot (T - \lambda_k I)^N$ to $\sum a_i v_i = 0$. The factors commute (T5) and $(T - \lambda_i I)^n v_i = 0$ (Prop. 70), so $Sv_i = 0$ for $i \neq k$.
3. What is left: $0 = a_k \prod_{i \neq k} (\lambda_k - \lambda_i)^n \, w$. The product and w are nonzero, so $a_k = 0$.

□

Two consequences.

- $G(\lambda, T) \cap G(\mu, T) = \{0\}$ for $\lambda \neq \mu$: a nonzero v in both would make the list v, v independent (Prop. 72). So a generalized eigenvector belongs to only one eigenvalue.
- If U is invariant, $G(\lambda, T|_U) = G(\lambda, T) \cap U$, since $(T|_U - \lambda I)^k u = (T - \lambda I)^k u$ for $u \in U$.

Theorem (Prop. 73 / LADR 8.9, 8.22: generalized eigenspace decomposition)
Over $\mathbb{C}$, if $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T, then $V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_m, T)$. In particular V has a basis of generalized eigenvectors of T.

The idea

(a) The $G(\lambda_i, T)$ together span V. (b) The sum is direct.

(b) is Prop. 72: a relation $u_1 + \cdots + u_m = 0$ with $u_i \in G(\lambda_i, T)$ and some $u_i \neq 0$ would be a dependence among generalized eigenvectors for distinct eigenvalues.

The splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$ with $U = \operatorname{range}(T - \lambda_1 I)^n$ invariant. U is smaller than V because $G(\lambda_1, T) \neq 0$.

Induct on $\dim V$. The theorem for $T|_U$ writes U as a sum of generalized eigenspaces of $T|_U$, and these lie inside those of T. The eigenvalue λ_1 does not come back in U, because $T - \lambda_1 I$ is injective on U.

Why $\mathbb{C}$: We need an eigenvalue λ_1 to start (T7). Over $\mathbb{R}$, the rotation $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ has no eigenvalues, so all its generalized eigenspaces are 0.

Example: $A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$: $G(1, A) = \operatorname{span}(e_1, e_2)$ (Ex. 61 below) and $G(3, A) = E(3, A) = \operatorname{span}\big((3, 2, 4)\big)$. So $\mathbb{C}^3 = G(1, A) \oplus G(3, A)$.

Proof. 1. Induction on n; $n = 1$ is clear. Take an eigenvalue λ_1 (T7).

2. Splitting result for $T - \lambda_1 I$: $V = G(\lambda_1, T) \oplus U$, $U = \operatorname{range}(T - \lambda_1 I)^n$. U is invariant (T5), and $\dim U < n$ since $G(\lambda_1, T) \neq 0$ (Prop. 71).
3. Eigenvalues of $T|_U$ are eigenvalues of T, but not λ_1 ($T - \lambda_1 I$ is injective on U). By induction $U = \bigoplus_{i \geq 2} G(\lambda_i, T|_U)$, and $G(\lambda_i, T|_U) \subseteq G(\lambda_i, T)$.
4. So the $G(\lambda_i, T)$ span V. Direct: if $u_1 + \cdots + u_m = 0$, $u_i \in G(\lambda_i, T)$, the nonzero u_i would be independent (Prop. 72); so all are 0.

□

On each $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent. So everything reduces to nilpotent operators (Ch. 15).

Definition (Def. 38)**. Algebraic multiplicity** of λ: $\dim G(\lambda, T)$. **Geometric multiplicity**: $\dim E(\lambda, T)$.

- geometric $\leq$ algebraic, since $E \subseteq G$.
- Over $\mathbb{C}$, the algebraic multiplicities add up to n (Prop. 73).
- Over $\mathbb{C}$, T is diagonalizable $\iff$ geometric = algebraic for every eigenvalue.
  Why. The sum $\sum E(\lambda_i, T)$ is direct (T7), and T is diagonalizable iff it is V, i.e. $\sum \dim E = n = \sum \dim G$. Since each $\dim E \leq \dim G$, the sums agree iff every term agrees.

Theorem (Prop. 74 / LADR 8.26, 8.29: Cayley–Hamilton)
Over $\mathbb{C}$, let $d_i = \dim G(\lambda_i, T)$ and $\chi_T(x) = \prod_i (x - \lambda_i)^{d_i}$ (the **characteristic polynomial**). Then $\chi_T(T) = 0$, so $m_T \mid \chi_T$. Also $\deg \chi_T = \sum d_i = n$ (Prop. 73), so $\deg m_T \leq n$.

The idea

$\chi_T(T)v = 0$ for every v.

$V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73), so it is enough to take v in one $G(\lambda_i, T)$. There, $T - \lambda_i I$ is nilpotent.

Which power of $T - \lambda_i I$ kills $G(\lambda_i, T)$? Prop. 70, applied inside $G(\lambda_i, T)$, a space of dimension d_i, gives the power d_i. That is why the exponents in χ_T are the d_i.

The factors of $\chi_T(T)$ commute, so for $v \in G(\lambda_i, T)$ let $(T - \lambda_i I)^{d_i}$ act first.

Proof. 1. $N = (T - \lambda_i I)|_{G(\lambda_i, T)}$ has $G(0, N) = G(\lambda_i, T)$, so $N^{d_i} = 0$ by Prop. 70 in that space.

2. For $v \in G(\lambda_i, T)$: $\chi_T(T)v = \prod_{j \neq i} (T - \lambda_j I)^{d_j} (T - \lambda_i I)^{d_i} v = 0$ (T5).
3. The G's span V, so $\chi_T(T) = 0$; then $m_T \mid \chi_T$ (T13(a)) and $\deg \chi_T = \sum d_i = n$.

□

- For a matrix, $\chi_A(x) = \det(xI - A)$ (stated in the notes; needs determinants).
- 2×2: $\chi_A = x^2 - (\operatorname{tr} A)x + \det A$.
  Why. Multiplying out gives $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq cI$, m_A has degree 2 and divides both this polynomial and χ_A, so all three agree. If $A = cI$, both are $(x-c)^2$.
- χ_T and m_T have the same roots; the exponents in m_T can be smaller.

Proposition (Exponents of m_T over $\mathbb{C}$, M6)
Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues and $d_i = \dim G(\lambda_i, T)$. Then $m_T = \prod_i (x - \lambda_i)^{s_i}$, where

$$s_i = \text{the least } k \text{ with } \operatorname{null}(T - \lambda_i I)^k = G(\lambda_i, T), \qquad 1 \leq s_i \leq d_i.$$

In particular $m_T \mid \chi_T$, and m_T has the same roots as χ_T.

The idea

The exponent of each $x - \lambda_i$ in m_T.

V splits into the invariant pieces $G(\lambda_i, T)$ (Prop. 73), and the minimal polynomial of a direct sum is the lcm of the pieces' minimal polynomials (T13(e)).

So compute m on each piece separately. On $G(\lambda_i, T)$ the operator $T - \lambda_i I$ is nilpotent, so the minimal polynomial there is a power of $x - \lambda_i$, and the right power is the first one that is 0 on the piece.

Proof. 1. $(T - \lambda_i I)^n = 0$ on $G(\lambda_i, T)$ (Prop. 70), so $m_{T|_{G(\lambda_i,T)}}$ divides $(x - \lambda_i)^n$. Its roots are roots of $(x - \lambda_i)^n$, so it is $(x - \lambda_i)^{s_i}$ for some s_i (over $\mathbb{C}$ it splits into linear factors, T6).

2. s_i is the least k with $(T - \lambda_i I)^k = 0$ on $G(\lambda_i, T)$, i.e. with $\operatorname{null}(T - \lambda_i I)^k \supseteq G(\lambda_i, T)$. Since $\operatorname{null}(T - \lambda_i I)^k \subseteq G(\lambda_i, T)$ always, this means equality.
3. By Prop. 73 and T13(e), $m_T = \operatorname{lcm}_i (x - \lambda_i)^{s_i} = \prod_i (x - \lambda_i)^{s_i}$ (the λ_i are distinct, so nothing is shared).
4. $s_i \geq 1$ since $G(\lambda_i, T) \neq 0$, and $s_i \leq d_i$ by Prop. 70 applied inside the d_i-dimensional $G(\lambda_i, T)$.

□

Computing m_T from null-space dimensions. For each eigenvalue λ, compute $d_k = \dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$ for $k = 1, 2, \ldots$. The first k at which d_k stops growing (equivalently, reaches the algebraic multiplicity) is the exponent of $x - \lambda$.

Example
$A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$ (Ex. 61). For $\lambda = 1$: $d_1 = 1$ and $d_2 = 2$, which is the number of 1s on the diagonal, so the exponent is 2. $\lambda = 3$ appears once, so its exponent is 1. So $m_A = (x-1)^2(x-3)$.

- With the multiplicities d_i known, the candidate test for m_T from Week 7 (candidates $\prod (x - \lambda_i)^{k_i}$, tested in increasing degree) needs only the candidates with $k_i \leq d_i$, and χ_T itself never needs testing (Cayley–Hamilton). In the Method 2 example, 1 has multiplicity 1, so $(x-3)(x-1)^2$ was never a candidate, and $(x-3)^2(x-1) = \chi_A$ needed no check.
- For 3×3 A (using $\chi_A = \det(xI - A)$, stated in the notes; determinants are Ch. 25): $\chi_A = x^3 - (\operatorname{tr} A)x^2 + c_2 x - \det A$, where c_2 is the sum of the three 2×2 minors on the diagonal (rows and columns $\{1,2\}, \{1,3\}, \{2,3\}$). This gives the eigenvalues and multiplicities for Method 2 when A is not triangular.

Proposition (LADR 8.31: diagonal of a triangular matrix)
Over $\mathbb{C}$, if $\mathcal{M}(T)$ is upper-triangular, then $\dim G(\lambda, T)$ is the number of times λ appears on the diagonal.

The idea

$\dim G(\lambda, T) = c_\lambda$, the number of times λ appears on the diagonal.

$\dim G(\lambda, T) = \dim \operatorname{null}(T - \lambda I)^n = n - \operatorname{rank}(T - \lambda I)^n$. The matrix of $(T - \lambda I)^n$ is upper-triangular with diagonal entries $(a_{jj} - \lambda)^n$, nonzero in exactly $n - c_\lambda$ places.

The obvious first attempt: Compute the rank exactly. This is hard: columns with a 0 on the diagonal can still be nonzero and add to the rank.

Settle for an inequality. The $n - c_\lambda$ columns with nonzero diagonal entries are independent (T8), so rank $\geq n - c_\lambda$, i.e. $\dim G(\lambda, T) \leq c_\lambda$. Now add over all λ: both sides add up to n, so none of the inequalities can be strict.

Pattern: "Prove $\leq$ for each term, then compare totals" also proves the test "diagonalizable $\iff$ geometric = algebraic" above.

Proof. 1. Let c_λ be the count. $\mathcal{M}((T - \lambda I)^n)$ is triangular with diagonal $(a_{jj} - \lambda)^n$, nonzero in $n - c_\lambda$ places.
2. Those columns are independent (T8), so rank $\geq n - c_\lambda$ and $\dim G(\lambda, T) \leq c_\lambda$ (Prop. 70, T1).
3. $n = \sum_\lambda \dim G(\lambda, T) \leq \sum_\lambda c_\lambda = n$ (Prop. 73), so every inequality is an equality.

□

Example (Computing $G(\lambda, A)$ (Ex. 61))

$A = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{smallmatrix}\right)$, $\lambda = 1$.

1. $\operatorname{null}(A - I) = \operatorname{span}(e_1)$, $\operatorname{null}(A - I)^2 = \operatorname{span}(e_1, e_2)$, $\operatorname{null}(A - I)^3 = \operatorname{span}(e_1, e_2)$.
2. Two consecutive null spaces agree, so the chain has stopped: $G(1, A) = \operatorname{span}(e_1, e_2)$.
3. Shortcut: 1 appears twice on the diagonal, so you can stop at dimension 2.

Shortcuts

- **Exponent 1 test.** The exponent of λ in m is 1 $\iff$ $\dim E(\lambda)$ equals the algebraic multiplicity $\iff$ $\operatorname{rank}(A - \lambda I) = n - d_\lambda$. One rank computation per eigenvalue settles diagonalizability.
- **Never test χ itself** when computing m: it always works (Cayley–Hamilton). If every smaller candidate fails, $m = \chi$.
- **Smaller power.** $G(\lambda, T) = \operatorname{null}(T - \lambda I)^d$, d the algebraic multiplicity, and even $\operatorname{null}(T - \lambda I)^k$ with k the exponent of $x - \lambda$ in m_T (exponents of m_T, above). Why: $T - \lambda I$ is nilpotent on the d-dimensional space $G(\lambda, T)$, so Prop. 70 inside it gives the power d.
- **Know the target dimension first.** For a triangular matrix, $\dim G(\lambda)$ is the number of λ's on the diagonal (LADR 8.31); in general it is the multiplicity of λ as a root of $\chi_A = \det(xI - A)$. Stop the chain $\dim \operatorname{null}(A - \lambda I)^k$ as soon as it reaches that number.
- **Last one by subtraction.** The algebraic multiplicities add up to n, so the last one is n minus the others. And if the $\dim G(\lambda_i)$ you found already add up to n, there are no other eigenvalues.
- **Multiplicity 1.** If the algebraic multiplicity is 1, then $G(\lambda) = E(\lambda)$ (since $1 \leq \dim E \leq \dim G = 1$).
- **One eigenvalue.** Over $\mathbb{C}$, if λ is the only eigenvalue, $G(\lambda, T) = V$ and $(T - \lambda I)^n = 0$ with no computation.
- **Geometric multiplicity by rank.** $\dim E(\lambda) = n - \operatorname{rank}(A - \lambda I)$, and $\dim \operatorname{null}(A - \lambda I)^k = n - \operatorname{rank}(A - \lambda I)^k$.
- **$G(\lambda)$ as a range.** Over $\mathbb{C}$, $G(\lambda, T) = \operatorname{range} \prod_{\mu \neq \lambda} (T - \mu I)^{d_\mu}$, and $\bigoplus_{\mu \neq \lambda} G(\mu, T)$ is the null space of that product. Useful when the other eigenvalues are simple: for the A of Ex. 61, $G(1, A) = \operatorname{range}(A - 3I) = \operatorname{span}(e_1, e_2)$. Why, with P the product and $H = \bigoplus_{\mu \neq \lambda} G(\mu, T)$:
  1. $P = 0$ on each $G(\mu, T)$, $\mu \neq \lambda$: let $(T - \mu I)^{d_\mu}$ act first (Prop. 74, step 1).
  2. P is injective on $G(\lambda, T)$: each $T - \mu I$ is, since a nonzero $v \in G(\lambda, T)$ with $Tv = \mu v$ would lie in $G(\lambda, T) \cap G(\mu, T) = 0$.
  3. $G(\lambda, T)$ is invariant (T5), so P maps it injectively into itself, hence onto itself (T2).
  4. $V = G(\lambda, T) \oplus H$ (Prop. 73) and $P = 0$ on H, so $\operatorname{range} P = G(\lambda, T)$ and $\operatorname{null} P = H$.
- **Commuting operators.** If $ST = TS$, then S maps $G(\lambda, T)$ into itself (G2): $(T - \lambda I)^n Sv = S(T - \lambda I)^n v = 0$.

Traps

- $G(\lambda, T)$ can be strictly bigger than $E(\lambda, T)$. Over $\mathbb{C}$ they agree for all λ exactly when T is diagonalizable.
- The exponents in m_T are at most the algebraic multiplicities, and they are not the geometric multiplicities: on $\mathbb{F}^4$ with $Ne_2 = e_1$ and $Ne_1 = Ne_3 = Ne_4 = 0$, $m_N = x^2$ while $\dim E(0) = 3$ and $\dim G(0) = 4$.
- A sum of generalized eigenvectors for different eigenvalues is not a generalized eigenvector (Prop. 72 makes the pieces independent).
- Prop. 73 needs eigenvalues: true over $\mathbb{C}$; over $\mathbb{R}$ only when m_T splits. (If m_T splits, every $T|_U$ with $U \neq 0$ invariant has an eigenvalue, because $m_{T|_U} \mid m_T$ (T13(d)) and its roots are eigenvalues (T13(b)); so the proof of Prop. 73 still works.)
- $V = \operatorname{null} T \oplus \operatorname{range} T$ can fail: for $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$. The splitting needs the power n.
- The power n always suffices; the chain often stops sooner.

§2.2 Nilpotent operators and Jordan form (Ch. 15)

The question. A complex operator is a direct sum of pieces $\lambda I + N$ with N nilpotent (Prop. 73), and λI looks the same in every basis. So we only need a good basis for a nilpotent N.

- Simplest nilpotent: a shift $v \mapsto Nv \mapsto \cdots \mapsto N^d v \mapsto 0$. In the basis $N^d v, \ldots, Nv, v$ its matrix has 1s just above the diagonal.
- The hope: every nilpotent operator is a direct sum of shifts.
- A basis of generalized eigenvectors is not enough: e_1, e_2, e_3 are generalized eigenvectors for 1 of I, $B = \left(\begin{smallmatrix} 3 & -1 & -1 \\ 2 & 0 & -1 \\ 2 & -1 & 0 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 3 & -2 & 0 \\ 1 & 1 & -1 \\ 0 & 2 & -1 \end{smallmatrix}\right)$, yet $\dim E(1) = 3, 2, 1$, so none are similar.

Definition (Def. 39)**.** N is **nilpotent** if $N^k = 0$ for some k; the least such k is the **nilpotency index**.

- $N^n = 0$: $V = G(0, N) = \operatorname{null} N^n$ (Prop. 70). So $m_N = x^k$, $k \leq n$.
- The only eigenvalue is 0 (if $V \neq 0$): $Nv = \lambda v \Rightarrow 0 = N^k v = \lambda^k v$. Over $\mathbb{C}$, conversely, only eigenvalue $0 \Rightarrow V = G(0, N)$ (Prop. 73).
- $N \neq 0$ is not diagonalizable: a diagonal matrix with all eigenvalues 0 is 0.
- Index $= \deg m_N \leq n$. HW8 P1: D on $\mathcal{P}_n(\mathbb{C})$ has index $n+1$, since $D^n x^n = n! \neq 0$ and $D^{n+1} = 0$.
- $(I - N)^{-1} = I + N + \cdots + N^{n-1}$: the product telescopes to $I - N^n = I$.

Proposition (LADR 8.18: nilpotent operators are strictly upper-triangular)
If N is nilpotent, then V has a basis in which $\mathcal{M}(N)$ is upper-triangular with every diagonal entry 0.

The idea

A basis $v_1, \ldots, v_n$ with $Nv_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$ for every j. That is exactly an upper-triangular matrix with zero diagonal.

If $N^k v = 0$, then $N^{k-1}(Nv) = 0$. So N maps $\operatorname{null} N^k$ into $\operatorname{null} N^{k-1}$.

List vectors by the least power of N that sends them to 0: a basis of $\operatorname{null} N$, extended to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$. A vector added at stage k is sent into $\operatorname{null} N^{k-1}$, which is spanned by vectors listed earlier.

Example: $N = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ ($Ne_1 = e_2$, $Ne_2 = 0$): $\operatorname{null} N = \operatorname{span}(e_2)$, so list e_2 first. In the basis e_2, e_1 the matrix is $\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$.

Proof. 1. Choose a basis of $\operatorname{null} N$, extend it to a basis of $\operatorname{null} N^2$, and so on up to $\operatorname{null} N^n = V$ (Props. 69, 70).
2. If v was added for $\operatorname{null} N^k$, then $N^{k-1}(Nv) = N^k v = 0$, so $Nv \in \operatorname{null} N^{k-1}$, the span of the earlier layers (for $k = 1$, $Nv = 0$).

3. So each basis vector is sent into the span of the vectors listed before it: column j has nonzero entries only above the diagonal.

□

Theorem (LADR 8.37: block diagonal form)

Over $\mathbb{C}$, let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T and $d_i = \dim G(\lambda_i, T)$. There is a basis in which $\mathcal{M}(T) = \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_m \end{pmatrix}$, where each A_i is a $d_i \times d_i$ upper-triangular matrix with every diagonal entry equal to λ_i.

The idea

A block diagonal matrix whose i-th block is upper-triangular with every diagonal entry λ_i.

Prop. 73 splits V into invariant pieces $G(\lambda_i, T)$, and invariant pieces give a block diagonal matrix (T10). So only the basis inside each piece is left to choose.

On $G(\lambda_i, T)$, $T = \lambda_i I + N_i$ with N_i nilpotent, and $\lambda_i I$ has the matrix $\lambda_i I$ in every basis. So choose the basis that makes N_i strictly upper-triangular (LADR 8.18); adding $\lambda_i I$ puts λ_i on the diagonal.

Proof. 1. $N_i = (T - \lambda_i I)|_{G(\lambda_i, T)}$ is nilpotent, since $G(\lambda_i, T) = \operatorname{null}(T - \lambda_i I)^n$ (Prop. 70). By LADR 8.18, $G(\lambda_i, T)$ has a basis in which $\mathcal{M}(N_i)$ is upper-triangular with zero diagonal.

2. In that basis $\mathcal{M}(T|_{G(\lambda_i, T)}) = \lambda_i I + \mathcal{M}(N_i)$: upper-triangular, diagonal entries λ_i, size d_i.

3. $V = \bigoplus_i G(\lambda_i, T)$ (Prop. 73) with each piece invariant, so concatenating the bases gives a block diagonal matrix (T10).

□

$$\mathcal{M}(T) = \left(\begin{array}{ccc|cc} \lambda_1 & * & * & 0 & 0 \\ 0 & \lambda_1 & * & 0 & 0 \\ 0 & 0 & \lambda_1 & 0 & 0 \\ \hline 0 & 0 & 0 & \lambda_2 & * \\ 0 & 0 & 0 & 0 & \lambda_2 \end{array}\right)$$

columns 1–3: a basis of $G(\lambda_1, T)$, $d_1 = 3$
columns 4–5: a basis of $G(\lambda_2, T)$, $d_2 = 2$

The zero blocks come from splitting V into invariant pieces (Prop. 73, T10). Each diagonal block is $\lambda_i I$ plus a strictly upper-triangular nilpotent part (LADR 8.18).

Proposition (Chain lemma)

If $N^d v \neq 0$ and $N^{d+1} v = 0$, then $v, Nv, \ldots, N^d v$ are independent.

The idea

From $\sum_{j=0}^{d} c_j N^j v = 0$, get every $c_j = 0$.

Applying N moves every term one step down the chain, and a term that is moved past $N^d v$ becomes 0.

Apply N^d: every term except $c_0 v$ is moved past $N^d v$ and becomes 0, leaving $c_0 N^d v = 0$, so $c_0 = 0$. Then N^{d-1} isolates c_1, and so on. Same idea as Prop. 72: kill every term but one.

Proof. Apply N^d to $\sum_j c_j N^j v = 0$: get $c_0 N^d v = 0$, so $c_0 = 0$. Then apply N^{d-1} to get $c_1 = 0$, and so on. □

Theorem (Prop. 75 / LADR 8.45: nilpotent normal form)

A nilpotent N has a basis made of chains $u, Nu, \ldots, N^k u$ ($N^{k+1} u = 0$). Listing each chain as $N^k u, \ldots, Nu, u$, the matrix $\mathcal{M}(N)$ has only 1s and 0s directly above the diagonal and 0 everywhere else.

The idea

The smallest case that shows everything. On $\mathbb{F}^3$ let $Ne_3 = e_2$, $Ne_2 = 0$, $Ne_1 = 0$. A chain is a list $u, Nu, N^2u, \dots$ stopped at its last nonzero vector. Here e_3, e_2 is a chain of length 2 and e_1 is a chain of length 1, and together they are a basis. In the order e_2, e_3, e_1 the matrix of N has a single 1 (from $Ne_3 = e_2$) and zeros elsewhere. The theorem says every nilpotent operator has a basis like this.

Where to look for the chains. Induction needs a smaller space that N maps into itself. $U = \operatorname{range} N$ works: it is invariant, and it is smaller than V because N is not injective (T2). In the example $U = \operatorname{span}(e_2)$, with chain basis e_2.

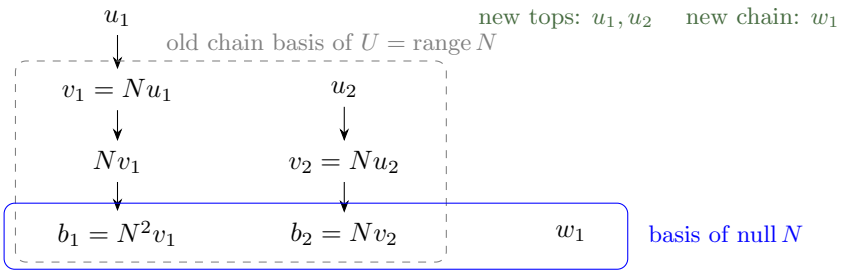
What a chain basis of U is missing. Compare it with what V needs.

(1) Every chain of U starts at a vector $v_i \in \operatorname{range} N$, so $v_i = Nu_i$ for some u_i. That u_i is not in the list yet. Put it on top; the chain becomes one longer. (Example: $e_2 = Ne_3$, so add e_3.)

(2) A vector that N sends to 0 but that is not in $\operatorname{range} N$ lies on no chain. (Example: e_1.) Such vectors must be added as chains of length 1.

Which vectors of $\operatorname{null} N$ are already in the list? Only the bottoms of the chains: the bottom is the one vector of a chain that N sends to 0. So extend the bottoms to a basis of $\operatorname{null} N$ by new vectors $w_1, \dots, w_l$.

What is left to check. That the new list has n vectors, and that it is independent. The count is rank–nullity: the old list has $\dim \operatorname{range} N$ vectors, and we added one top per chain (r of them) and l vectors w, where $r + l = \dim \operatorname{null} N$. Independence is the step that needs care. The plan: apply N to a relation. That removes the w's and the bottoms and turns everything else into vectors of the old chain basis of U, which we know is independent.



Two chains of U (inside the dashed box) with $d_1 = 2$, $d_2 = 1$. Each gets a new top u_i with $Nu_i = v_i$. The bottoms b_1, b_2 are extended by w_1 to a basis of $\operatorname{null} N$ (bottom row). Arrows are N; the bottom row is sent to 0.

Proof. 1. If $N = 0$, any basis works (chains of length 1). Otherwise $U = \operatorname{range} N$ is invariant with $0 < \dim U < n$. By induction U has a chain basis $v_i, Nv_i, \dots, N^{d_i}v_i$ for $i = 1, \dots, r$, with $N^{d_i+1}v_i = 0$. Write $b_i = N^{d_i}v_i$ for the bottoms.

2. *The bottoms are a basis of* $\operatorname{null} N \cap U$. They lie in $\operatorname{null} N$ and in U, and they are independent (part of a basis). They span: take $x \in U$ with $Nx = 0$ and write $x = \sum_{i,j} c_{ij} N^j v_i$. Then $0 = Nx = \sum_{j<d_i} c_{ij} N^{j+1} v_i$ (the terms with $j = d_i$ are 0). The vectors $N^{j+1}v_i$ with $j < d_i$ are different members of the chain basis, so $c_{ij} = 0$ for $j < d_i$, and $x = \sum_i c_{id_i} b_i$.

3. Choose u_i with $Nu_i = v_i$ (possible since $v_i \in \operatorname{range} N$). Extend $b_1, \dots, b_r$ by $w_1, \dots, w_l$ to a basis of $\operatorname{null} N$. So $\dim \operatorname{null} N = r + l$.

4. *The new list* is $u_i, v_i, Nv_i, \dots, b_i$ for each i, together with $w_1, \dots, w_l$. *Count:* $\sum_i (d_i + 2) + l = \sum_i (d_i + 1) + r + l = \dim \operatorname{range} N + \dim \operatorname{null} N = n$ (T1).

5. *Independence, first half.* Suppose $\sum_i a_i u_i + \sum_{i,j} c_{ij} N^j v_i + \sum_k e_k w_k = 0$. Apply N. We have $Nu_i = v_i$; $N(N^j v_i) = N^{j+1} v_i$, which is 0 for $j = d_i$; and $Nw_k = 0$. So $\sum_i a_i v_i + \sum_{j<d_i} c_{ij} N^{j+1} v_i = 0$. The vectors here are different members of the chain basis of U (the tops v_i, and the vectors below them). So every $a_i = 0$, and $c_{ij} = 0$ for $j < d_i$.

6. *Independence, second half.* What is left of the relation is $\sum_i c_{id_i} b_i + \sum_k e_k w_k = 0$. The b_i and w_k together are a basis of $\operatorname{null} N$ (step 3), so these coefficients are 0 too. This is the reason the w's were chosen to extend the bottoms: a w that was a combination of chain vectors would, by step 5, be a combination of the bottoms, and then bottoms and w's would be dependent.

7. So the new list is n independent vectors, a basis (T3), and it consists of chains. Listing a chain as $N^k u, \dots, Nu, u$, N sends each vector to the one before it and the first to 0. That is a column with a single 1 directly above the diagonal, or a zero column.

□

Definition (Def. 40)**. Jordan block**: $J_k(\lambda) = \lambda I + N_k$ ($k \times k$, 1s on the superdiagonal), so $Je_1 = \lambda e_1$, $Je_j = \lambda e_j + e_{j-1}$. **Jordan normal form**: block diagonal with Jordan blocks. **Jordan basis**: a basis giving Jordan form.

Example (HW8 P1). D on $\mathcal{P}_n(\mathbb{C})$ is the single block $J_{n+1}(0)$ in the basis $1, x, \frac{x^2}{2!}, \dots, \frac{x^n}{n!}$: $D\frac{x^k}{k!} = \frac{x^{k-1}}{(k-1)!}$.

Theorem (Prop. 76 / LADR 8.46: Jordan normal form)
Every operator on a finite-dimensional complex space has a Jordan basis.

The idea

A basis in which T is block diagonal with Jordan blocks.

Prop. 73 splits V into invariant pieces $G(\lambda_i, T)$, so the matrix is block diagonal (T10). On each piece $T = \lambda_i I + N_i$ with N_i nilpotent, and $\lambda_i I$ has the same matrix in every basis.

Use the chain basis of N_i (Prop. 75) in each piece: N_i becomes a sum of blocks N_k, so T becomes a sum of blocks $\lambda_i I + N_k = J_k(\lambda_i)$. This is the proof of LADR 8.37 with the chain basis in place of the triangular one.

Proof. On each invariant $G(\lambda_i, T)$, $N_i = (T - \lambda_i I)|_G$ is nilpotent. Prop. 75 gives a basis where N_i is a sum of blocks N_k, so $T = \lambda_i I + N_i$ is a sum of blocks $J_k(\lambda_i)$. Concatenate the bases (T10). □

Compared with LADR 8.37. Both start by splitting V into the $G(\lambda_i, T)$. LADR 8.37 only makes each block upper-triangular with λ_i on the diagonal; Jordan form also makes every entry above the diagonal 0, except the entries directly above it, which are 0 or 1. That needs the chain construction of Prop. 75.

Reading the Jordan form. For each eigenvalue λ:

- number of blocks for λ = geometric multiplicity $\dim E(\lambda, T)$;
- total size of those blocks = algebraic multiplicity $\dim G(\lambda, T)$;
- largest block for λ = exponent of $(x - \lambda)$ in m_T;
- $\chi_T = \prod (x - \lambda)^{\text{alg. mult.}}$, $m_T = \prod (x - \lambda)^{\text{largest block}}$;
- diagonalizable $\iff$ all blocks 1×1 $\iff$ m_T has no repeated root.

Why.

1. One block $J_k(\lambda)$: $J_k(\lambda) - \lambda I = N_k$ moves each basis vector one place ($e_j \mapsto e_{j-1}$, $e_1 \mapsto 0$), so N_k^j moves it j places and $\dim \operatorname{null} N_k^j = \min(j, k)$. Hence $E = \operatorname{span}(e_1)$ (one block, dimension 1), G is the whole block (size k), and $m = \chi = (x - \lambda)^k$.
2. A block $J_j(\mu)$ with $\mu \neq \lambda$ adds nothing to $\operatorname{null}(T - \lambda I)^k$: $(J_j(\mu) - \lambda I) = (\mu - \lambda)I + N_j$ is invertible (a nilpotent has only the eigenvalue 0, so $N_j v = (\lambda - \mu)v$ forces $v = 0$), and so are its powers.
3. For the whole matrix, null spaces add over the blocks (T10) and m is the lcm of the blocks' minimal polynomials (T13(e)). Adding up gives each line of the box.

Theorem (Prop. 77: counting blocks)
Over $\mathbb{C}$, let $d_k = \dim \operatorname{null}(T - \lambda I)^k$ ($d_0 = 0$). For $k \geq 1$:

- number of blocks for λ of size $\geq k = d_k - d_{k-1}$;
- number of size exactly $k = 2d_k - d_{k-1} - d_{k+1}$.

So the Jordan form is unique up to order, and two complex matrices are similar iff they have the same Jordan form.

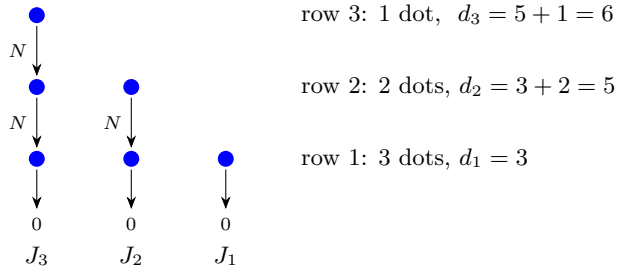
The idea

The block sizes in terms of numbers that do not depend on a basis. Then two Jordan bases must give the same blocks.

$d_k = \dim \operatorname{null}(T - \lambda I)^k$ is defined from T alone. In a Jordan basis the matrix is block diagonal, so null spaces split block by block (T10), and d_k is a sum over the blocks.

Compute one block. $J_j(\lambda) - \lambda I = N_j$ is a shift, so $\dim \operatorname{null} N_j^k = \min(k, j)$: it goes up by 1 at each step until $k = j$ and then stays at j. So from $k-1$ to k, exactly the blocks of size $\geq k$ add 1. Blocks for other eigenvalues add 0.

Example: Blocks $J_3(0), J_1(0)$: $d_1, d_2, d_3 = 2, 3, 4$. So 2 blocks of size ≥ 1, 1 of size ≥ 2, 1 of size ≥ 3.



A nilpotent N with blocks $J_3(0), J_2(0), J_1(0)$. Each column is one chain, read downward; the bottom dot is an eigenvector. N^k sends exactly the dots in the bottom k rows to 0, so d_k is the number of dots in rows $1, \ldots, k$, and $d_k - d_{k-1}$ is the number of dots in row k, i.e. the number of blocks of size $\geq k$.

Proof. 1. In a Jordan basis, $d_k = \sum_{\text{blocks}} \dim \operatorname{null}(J - \lambda I)^k$. Blocks for $\mu \neq \lambda$ contribute 0; $J_j(\lambda)$ contributes $\min(k, j)$.

2. $\min(k, j) - \min(k-1, j)$ is 1 if $j \geq k$, else 0. Summing: $d_k - d_{k-1} = \#\{\text{blocks with } j \geq k\}$. Exactly k: $(d_k - d_{k-1}) - (d_{k+1} - d_k)$.

3. Similar matrices are one operator in two bases (T11), so they have the same d_k and the same Jordan form. Conversely, matrices with the same Jordan form are both similar to it.

□

Example (Jordan form from null-space dimensions (Ex. 63))

$T \in \mathcal{L}(\mathbb{C}^8)$ with eigenvalues $1, 2$. Method: list $d_0, d_1, \ldots$, take differences, then differences of differences.

λ	d_0	d_1	d_2	d_3	d_4	blocks
1	0	2	4	4		sizes ≥ 1: 2, ≥ 2: 2, ≥ 3: 0 $\Rightarrow J_2(1), J_2(1)$
2	0	2	3	4	4	sizes ≥ 1: 2, ≥ 2: 1, ≥ 3: 1, ≥ 4: 0 $\Rightarrow J_3(2), J_1(2)$

($d_4 = 4$ for $\lambda = 2$ because $\dim G(2, T) = 8 - 4$.) Answer: $J_2(1) \oplus J_2(1) \oplus J_3(2) \oplus J_1(2)$.

Example (Finding a Jordan basis (Ex. 62))

$A = \left(\begin{smallmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right)$: only eigenvalue 1, $d_1, d_2, d_3 = 1, 2, 3$, so one block $J_3(1)$.

1. Pick the top: $v_3 \in \operatorname{null}(A-I)^3$, $v_3 \notin \operatorname{null}(A-I)^2$. Here $v_3 = e_3$.
2. Go down: $v_2 = (A-I)v_3 = (3,2,0)$, $v_1 = (A-I)v_2 = (4,0,0)$ (an eigenvector).
3. Check: $Av_1 = v_1$, $Av_2 = v_2 + v_1$, $Av_3 = v_3 + v_2$. That is $J_3(1)$.

Why start at the top: the top vector determines the whole chain, and a top outside $\operatorname{null}(A-\lambda I)^{k-1}$ guarantees full length k (chain lemma).

Several blocks: get the block sizes first (Prop. 77). For each block of size j, largest first, pick a top $v \in \operatorname{null}(A-\lambda I)^j \setminus \operatorname{null}(A-\lambda I)^{j-1}$ and keep its chain only if its bottom $(A-\lambda I)^{j-1}v$ is independent of the bottoms already chosen. (Independent of the vectors so far is not enough: in $J_3(0) \oplus J_2(0)$ with chains e_1, e_2, e_3 and f_1, f_2, the top $f_1 + e_2$ gives a chain ending at e_1 again.)

Powers. $J = \lambda I + N$ with λI and N commuting, so the binomial theorem applies and $N^j = 0$ for $j \geq k$:

$$J_k(\lambda)^m = \sum_{j=0}^{\min(m,k-1)} \binom{m}{j} \lambda^{m-j} N^j, \qquad J_3(\lambda)^m = \begin{pmatrix} \lambda^m & m\lambda^{m-1} & \binom{m}{2}\lambda^{m-2} \\ 0 & \lambda^m & m\lambda^{m-1} \\ 0 & 0 & \lambda^m \end{pmatrix}.$$

- Example: $J_3(1)^{100}$ has 100 and 4950 above the diagonal.
- Any Jordan matrix is $J = D + N$, D diagonal, N nilpotent, $DN = ND$ (on each block $D = \lambda I$); this is the Jordan–Chevalley decomposition. If J is invertible, also $J = DU$ with $U = I + D^{-1}N$, $DU = UD$.
- **Square roots (LADR 8.39–8.41).** $I + N$ has a square root: choose a_j with $(I + a_1 N + \cdots + a_{n-1}N^{n-1})^2 = I + N$ by matching coefficients ($2a_1 = 1$, $2a_2 + a_1^2 = 0$, …); powers N^n and up vanish. This is the series $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$ cut off. Over $\mathbb{C}$, every invertible operator has a square root (apply this to $\lambda_i(I + N_i/\lambda_i)$ on each $G(\lambda_i, T)$).

Traps

- χ_T and m_T do not fix the Jordan form: $J_2(0) \oplus J_2(0)$ and $J_2(0) \oplus J_1(0) \oplus J_1(0)$ both have $\chi = x^4$, $m = x^2$.
- Adding the number of blocks is still not enough: $J_3 \oplus J_3 \oplus J_1$ and $J_3 \oplus J_2 \oplus J_2$ (all for 0) agree in χ, m and block count, but $d_2 = 5$ vs. 6. You need all the d_k.
- Order each chain from the bottom (the eigenvector) up; reversed, the 1s go below the diagonal.
- Jordan form needs $\mathbb{C}$ (or split m_T): the rotation of $\mathbb{R}^2$ has none over $\mathbb{R}$.
- The exponent of λ in m_T is the size of the largest block for λ, not the number of blocks (that is $\dim E(\lambda, T)$).

§3 Problem set

§3.1 Homework 8 and extra problems

HW8 P1. Let $D\colon \mathcal{P}_n(\mathbb{C}) \to \mathcal{P}_n(\mathbb{C})$ be differentiation.

(a) Prove that D is nilpotent and determine its nilpotency index.
(b) Determine the geometric and algebraic multiplicities of the eigenvalue 0.
(c) Find a basis with respect to which the matrix of D is a single Jordan block.

HW8 P2. Construct explicit matrices having the following spectral data.

(a) Eigenvalue 2 has algebraic multiplicity 3 and geometric multiplicity 1; eigenvalue -1 has algebraic multiplicity 2 and geometric multiplicity 2.
(b) The same algebraic multiplicities as in (a), but the geometric multiplicity of 2 is 2.

For each matrix, give its minimal polynomial.

HW8 P3. Let $J = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) Compute $\operatorname{null}(J - I)$, $\operatorname{null}((J - I)^2)$, and $\operatorname{null}((J - I)^3)$.
(b) Determine the generalized eigenspace $G(1, J)$ and the geometric and algebraic multiplicities of 1.
(c) Compute J^m for every positive integer m.

HW8 P4. Let N be nilpotent with Jordan form consisting of one block of size 3 and one block of size 2.

(a) Compute $\dim \operatorname{null} N^k$ for $k = 0, 1, 2, 3, 4$.
(b) Suppose instead that you are only told $\dim \operatorname{null} N = 2$, $\dim \operatorname{null} N^2 = 4$, $\dim \operatorname{null} N^3 = 5$. Explain how these numbers force the same two block sizes.

HW8 P5. (optional) Let $T \in \mathcal{L}(\mathbb{C}^6)$ have eigenvalues 1 and 2, and suppose

$$\dim \operatorname{null}(T - I) = 1, \quad \dim \operatorname{null}((T - I)^2) = 2, \quad \dim \operatorname{null}((T - I)^3) = 2,$$

while

$$\dim \operatorname{null}(T - 2I) = 2, \quad \dim \operatorname{null}((T - 2I)^2) = 3, \quad \dim \operatorname{null}((T - 2I)^3) = 4, \quad \dim \operatorname{null}((T - 2I)^4) = 4.$$

Determine the Jordan normal form of T, up to reordering of the blocks.

HW8 P6. (optional) Suppose an operator has Jordan blocks $J_3(4)$, $J_2(4)$, $J_1(-1)$. Determine:

(a) its characteristic polynomial;
(b) its minimal polynomial;
(c) the geometric and algebraic multiplicities of each eigenvalue;
(d) whether it is diagonalizable.

HW8 P7. (optional) Prove that for a nilpotent operator N, $\dim \operatorname{null} N^k - \dim \operatorname{null} N^{k-1}$ is the number of Jordan blocks of size at least k.

G1. Let $A = \begin{pmatrix} 1 & 1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ act on $\mathbb{C}^3$. Find the eigenvalues, $E(2, A)$, $G(2, A)$, the Jordan form, a Jordan basis, and m_A.

G2. Let V be finite-dimensional and $S, T \in \mathcal{L}(V)$ with $ST = TS$. Prove that $G(\lambda, T)$ is invariant under S.

G3.

(a) Let N be nilpotent on a finite-dimensional $V \neq \{0\}$. Prove that $\lambda I + N$ is invertible if and only if $\lambda \neq 0$.
(b) Let V be a finite-dimensional complex vector space and suppose $T \in \mathcal{L}(V)$ has only one eigenvalue λ. Prove that $T - \lambda I$ is nilpotent.

G4. Find all possible Jordan forms of $T \in \mathcal{L}(\mathbb{C}^4)$ with $\chi_T = (x - 2)^4$ and $m_T = (x - 2)^2$. Which single number tells them apart?

G5. Let V be a finite-dimensional complex vector space and $T \in \mathcal{L}(V)$. Prove that if every generalized eigenvector of T is an eigenvector, then T is diagonalizable.